

Atharva Agrawal

+91 9617710304

✉ agrawalatharva2004@gmail.com

🌐 LinkedIn

🐙 GitHub

Portfolio

Education

Indian Institute of Information Technology, Dharwad

Aug 2023 – Present

B.Tech in Data Science & Artificial Intelligence CPI: 8.82 — Minor in Cybersecurity

Relevant Coursework: DSA, Object Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering, Cryptography, DevSecOps.

Experience

Hindustan Aeronautics Limited

June 2026 – Present

Full Stack Developer Intern (Remote)

- Built a full-stack procurement portal (React 18 + Node/Express) with **6 role-based screens, JWT auth, and an 8-stage payment state machine** — all money/GST/liquidated-damages math computed server-side across 6 user roles.
- Developed 2 SQLite-backed modules — an e-file noting workflow with custody-based access control and 4-level document classification, and a contract engine generating documents from a **71-clause × 8-type matrix** with **SHA-256** tamper-verification and QR signing.
- Integrated an AI pipeline (**local Qwen 2.5 SLM**) into the web app to auto-generate 10-stage procurement notes, surfaced read-only through a config-driven document viewer.

OMS - Office of Pralhad Joshi (Union Minister, Govt of India)

Dec 2025 – June 2026

Full Stack Developer Intern

- Architected and deployed a production-grade **Office Management System (OMS)** from scratch on the **Zoho platform**, automating grievance handling, visitor tracking, and administrative workflows across **6+ operational modules**.
- Designed and implemented an automated **CI/CD pipeline** for build, testing, and deployment workflows, enabling faster release cycles, reducing deployment overhead, and ensuring reliable production deployments.
- Optimized backend queries, API execution, and frontend rendering pipelines, reducing average application latency from **7–8 seconds to nearly 1 second (~85% performance improvement)** and significantly enhancing user experience.
- Built and integrated **15+ RESTful APIs** with **Node.js, Express.js, and Prisma ORM** along with responsive dashboards using **React**, improving internal workflow efficiency and reducing manual processing effort by **60%**.
- Engineered secure **RBAC-based authentication and authorization** supporting **multiple user roles** and securing access across **10+ protected API endpoints** with verification and document-generation workflows.

Projects

Kubernetes Attack Path Analyzer



Kubernetes, Docker, React, Python, FastAPI

- Developed a production-grade **Kubernetes Attack Path Analyzer** leveraging **graph algorithms (BFS, DFS, Dijkstra, and Betweenness Centrality)** to detect **46+ potential attack paths**, privilege escalation loops, and critical infrastructure chokepoints in Kubernetes clusters.
- Engineered a scalable full-stack security analytics platform using **FastAPI, React 18, NetworkX, Docker, and Cytoscape.js**, capable of analyzing **500-node Kubernetes graphs in under 1 second** with optimized in-memory graph processing.
- Built and integrated **RESTful security APIs** for attack-path tracing, blast-radius analysis, privilege escalation detection, and AI-generated remediation reporting, achieving **85%+ automated test coverage** across algorithms and end-to-end workflows.
- Designed an AI-powered kill-chain reporting engine using **Gemini 2.0 Flash** to convert complex graph-security analysis into actionable executive-level threat narratives and remediation recommendations for cloud infrastructure security teams.

AI Scholar – Research Paper Question Answering System



React Native, Express.js, FastAPI, MongoDB, ChromaDB

- Built an accessible mobile learning platform that transforms research papers into a **conversational, voice-enabled AI experience**, supporting **natural language queries** and **audio responses**.
- Engineered a **Retrieval-Augmented Generation (RAG) pipeline** sourcing context from a **vector database** to ensure **accurate, citation-backed answers** with minimal hallucinations.
- Delivered **secure authentication, persistent history, streaming responses, Voice-to-Text/Text-to-Speech, dark mode**, and an **automated PDF ingestion engine** for dynamic knowledge expansion.

Technical Skills

Programming Languages: C, C++, TypeScript, JavaScript

Backend: Node.js, Express.js, FastAPI, REST APIs

Frontend: React, React Native, Tailwind CSS

Databases: MySQL, MongoDB, PostgreSQL, Prisma ORM

Tools: Docker, Jenkins, Kubernetes, AWS, Git, Linux, Firebase

Achievements

- **LeetCode:** 1600+ contest rating (Solved 500+ questions)
- **HackerRank:** 5-star in C++, SQL Certified
- **CodeChef:** 3-Star Max(1627)